Attention Is All You Need - NIPS 2017

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The dominant competitive models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train.

Encoder and Decoder Stacks: The encoder is composed of a stack of N = 6 identical layers. Each layer has two sub-layers: Multi-Head Self-Attention and position-wise fully connected feed-forward network.